



# 清华大学

## 习题一

$$\begin{aligned} 1. a) & xyz + x'y + xy z' \\ &= xy(z+z') + x'y \\ &= (x+x')y = y \end{aligned}$$

$$\begin{aligned} b) & (a+b+c')(a'b'+c) \\ &= a'b'c' + (a+b)c \end{aligned}$$

$$\begin{aligned} c) & a'b'c + abc' + abc + a'bc' \\ &= bc + bc' = b \end{aligned}$$

$$\begin{aligned} d) & (BC' + A'D)(AB' + CD') \\ &= 0 \end{aligned}$$

$$2. a) \text{ 解: } [(a+c)(a+b')(a'+b+c')] = a'c' + a'b + ab'c$$

$$b) \text{ 解: } [z+z'(v'w+xy)] = z'(z+(v+w')(x'+y')) = z'(v+w')(x'+y')$$

$$3. F(A, B, C, D) = B'D + A'D + BD$$

$$\text{解: } F(A, B, C, D) = (B' + A' + B)D = D$$

$$= ABCD + A'BCD + AB'CD + ABC'D + A'B'CD + AB'C'D + A'BC'D + A'B'C'D$$

$$= (A+B+C+D)(A'+B+C+D)(A+B'+C+D)(A+B+C'+D)(A'+B'+C+D)(A+B'+C'+D)(A'+B+C'+D)(A'+B'+C'+D)$$

$$4. a) (u+xw)(x+u'w)$$

$$= xuw + xu'w' + xu'w$$

$$= (x+u+w)(x+u'+w)(x+u+w)(x+u'+w)(x'+u+w)$$

$$b) x' + x(x+y')(y+z')$$

$$= xyz' + xyz + xy'z' + x'yz + x'y'z + x'yz' + x'y'z'$$

$$= x' + y + z'$$

5. a)

| x | y | z | w | F |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

c)  $F = xy'z + x'y'z + w'xy + wx'y + wxy$

$= y'z + wx'y + xy$

$= y'z + (wx' + x)y$

$= y'z + (x+w)y$

